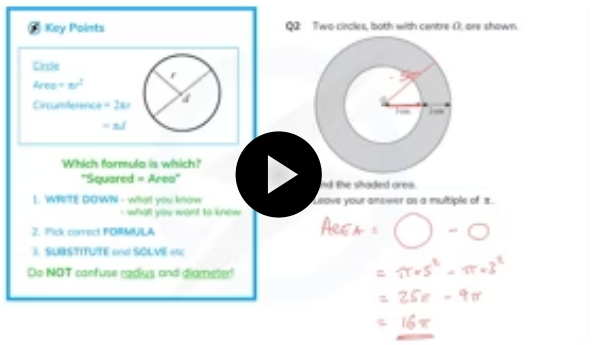
Area & Circumference of Circles

Exam code: 0607

Download PDF

Written by: Naomi C
Reviewed by: Dan Finlay
Updated on 19 January 2026



Did this video help you? Yes No

Area & Circumference

What are the properties of a circle?

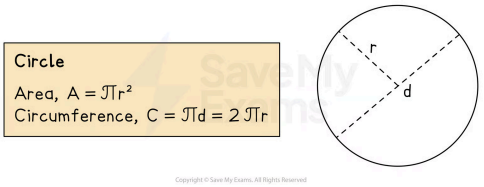
- A **circle** is a shape that is made up of all the points on a 2D plane that are **equidistant** from a single point
  - Equidistant means the same distance
- The **circumference** of a circle is its **perimeter**
- The **diameter**, ***d***, of a circle is **twice** its **radius**, ***r***
- π **(pi)** is the number (3.14159 ...) that is the ratio between a circle's **diameter** and its **circumference**
  - You may be asked to give an answer to a question as an 'exact value' or 'in terms of π'

What circle formulae do I need to know?

- The **formulae** for the **area** and **circumference** of a circle are **given to you in your exam**
  - However, it can be useful to remember them!
- The **area of a circle** can be calculated using the formula:
  - $A = \pi r^2$
- The **circumference of a circle** can be calculated using either of the following two formulae:

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- Working with circle formulae is just like working with any other formula:
  - **Write down** what you know (or what you want to know)
  - Pick the correct **formula**
  - **Substitute** the values in and **solve**

How do I find the circumference of a circle?

- Identify the **diameter**
  - This is **double** the length of the **radius**
- **Multiply** the **diameter** by **π**

How do I find the area of a circle?

- Identify the **radius**
  - This is **half** the length of the **diameter**
- **Square** the **radius**
- **Multiply** the **radius squared** by **π**

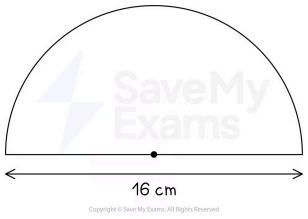
Examiner Tips and Tricks

- Remember that area is always measured in square units (cm^2, m^2, ... etc.) and circumference is always measured in units of a single length (cm, m, ... etc.)

Worked Example

Find the area and perimeter of the semicircle shown in the diagram.

Give your answers in terms of π.



the same diameter, so begin by finding the area of the full circle

Find the radius by dividing the diameter by 2

$$r = \frac{16}{2} = 8 \text{ cm}$$

Substitute this into the formula for the area of a circle $A = \pi r^2$.
Leave your answer in terms of π, (this just means do not multiply by π on your calculator)

$$A_{\text{full circle}} = \pi(8)^2 = 64\pi$$

Find the area of the semicircle by dividing the full area by 2

$$A_{\text{semicircle}} = \frac{1}{2}(64\pi) = \frac{64}{2}\pi = 32\pi$$

Area = 32π cm²

The **perimeter** of the semicircle is made up of both the arc of the circle (half of the circumference) and the diameter of the semicircle

Find the full length of the circumference of the circle using the formula

$$C = 2\pi r \text{ (or } C = \pi d)$$

Substitute the radius = 8 cm into the formula
Again, leave your answer in terms of π

$$C = 2\pi(8) = 16\pi$$

Find the length of the arc (the curved part of the perimeter of the semicircle) by dividing the full area by 2

$$\text{Curved length} = \frac{1}{2}(16\pi) = \frac{16}{2}\pi = 8\pi$$

Find the full perimeter by adding this to the length of the diameter of the circle

Perimeter = (8π+16) cm

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder →

Test yourself

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.